

Accelerated Volumetric Compression without Hierarchies: A Fourier Feature Based Implicit Neural Representation Approach

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Motivation

Volumetric data is large, slow to process, and costly to store, especially in high-resolution simulations or scans. Solutions like NeuralVDB tackles this with sparse, hierarchical neural structures. We take a different approach: a structure free implicit neural representation using Fourier features, now accelerated by selective voxel sampling.

Methods

We train a coordinate-based MLP with Fourier Feature mapping to represent 3D scalar fields. Instead of training on all voxels, we use an active voxel mask (AVM) to shrink down the training dataset and thus speed up the training.

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Main findings

Training on AVM only can lead to quality degradation and artifacts, as the INR does not have much context. So we enlarged the AVM by dilation of ℓ to help preserve reconstruction quality compared to training on the whole bounding box (BBX), while still cutting training time drastically.

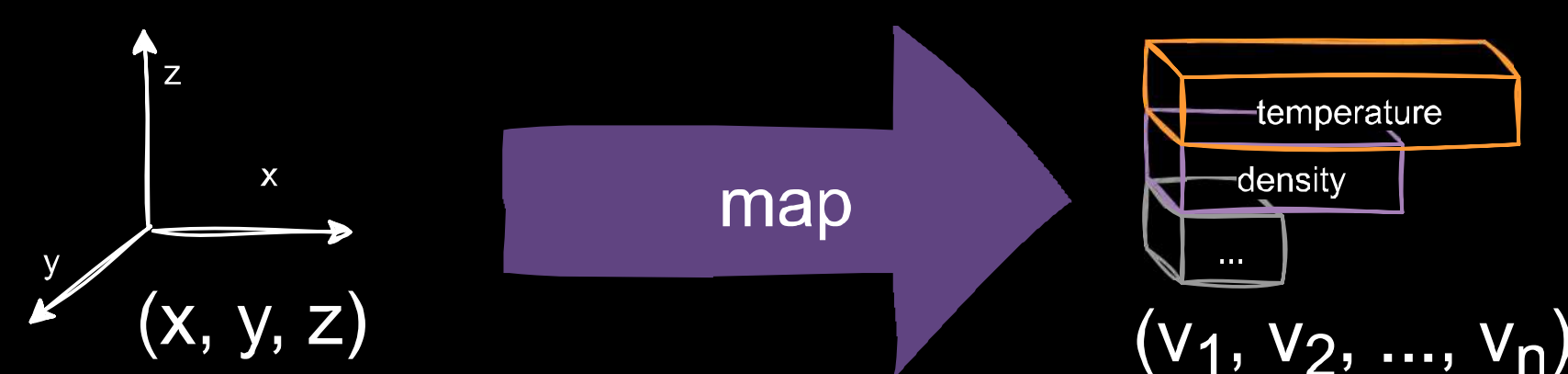
By doing so, we found out that very small dilation (like $\ell = 1$) still produces some artifacts, especially on sharp edges. More aggressive dilation helped, in this case $\ell = 5$. By saving weights only, we achieved 14× compression, and by training only on dilated AVM, we sped up training by 63.7%.

Variation	PSNR (dB)	SSIM	Time (s)
AVM	15.47	0.536	554
$\ell = 1$	29.11	0.917	585
$\ell = 2$	28.41	0.916	576
$\ell = 5$	32.01	0.940	646
$\ell = 10$	32.21	0.947	693
BBX	32.60	0.948	1780

The AVM, $\ell=5$ and BBX variations are visualized as slices in the "Faster training" part. The BBX and $\ell=5$ variations are also visualized alongside the original (biggest visual on the poster.)

Implicit Neural Representation (INR)

Store less values. INRs store only the weights of a trained network that maps coordinates to data values, drastically reducing storage.



Faster training

Volumetric data often includes large background regions with zero or near-zero values. Skipping them speeds up training, but using only active voxels can degrade quality. We apply dilation to preserve context and maintain reconstruction fidelity.

Presented dataset resembles a gas burner and contains two scalar grids (temperature & density) each with a resolution of $239 \times 239 \times 387$, totaling over 44 million values. Stored in VDB format, the volume takes up 43 MB on disk. Images are visualizing only density grid, but the INR is trained on all the variables.

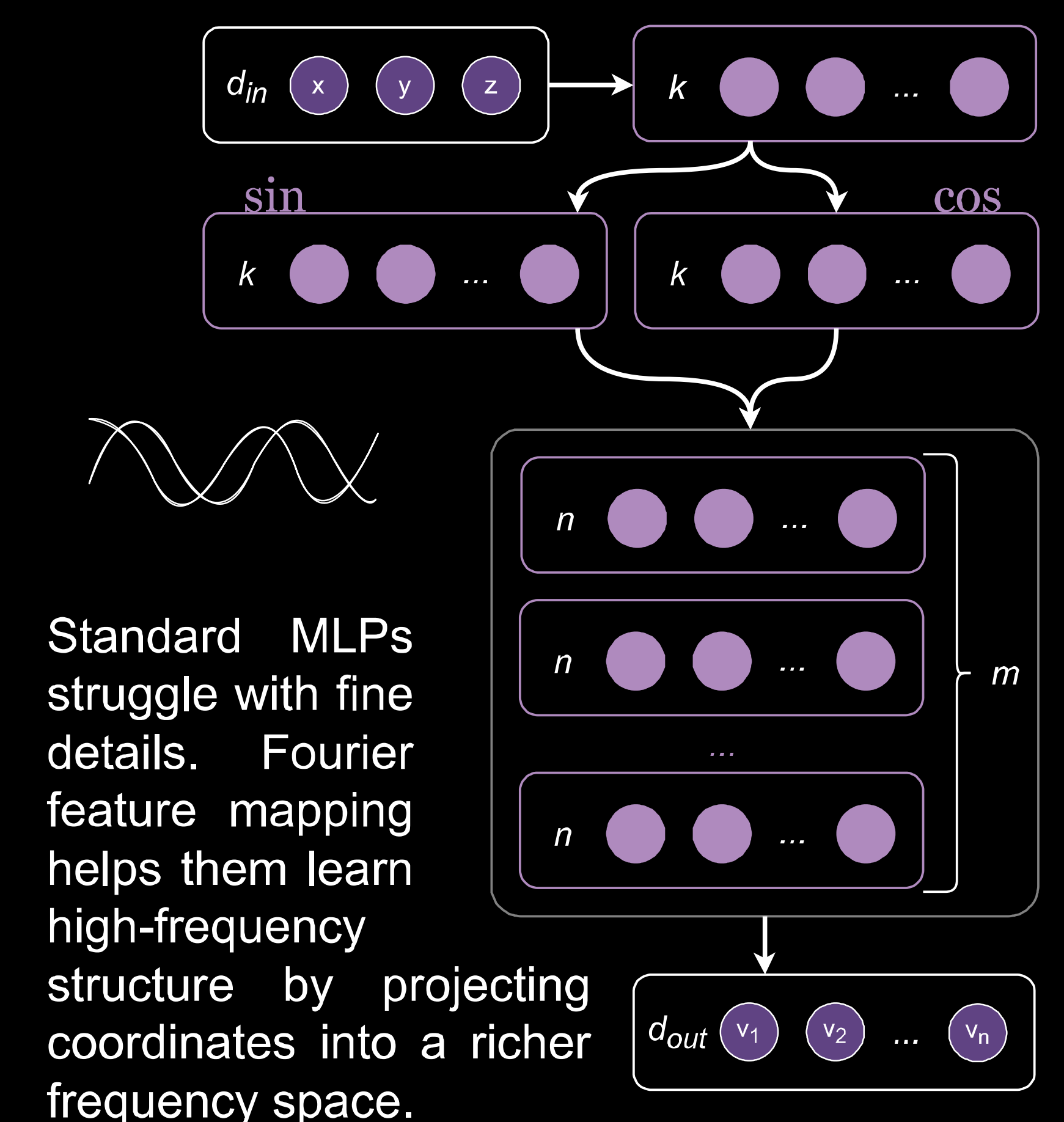
Limitations & future work

Optimal dilation depends on dataset geometry. Aggressive or minimal dilation may under- or over- approximate boundaries. The method currently uses a fully connected MLP. Alternative architectures (SIREN, convolutional INRs) could further improve speed or quality. Future directions include network pruning, adaptive sampling strategies, and tuning for different domain-specific volumetric tasks.

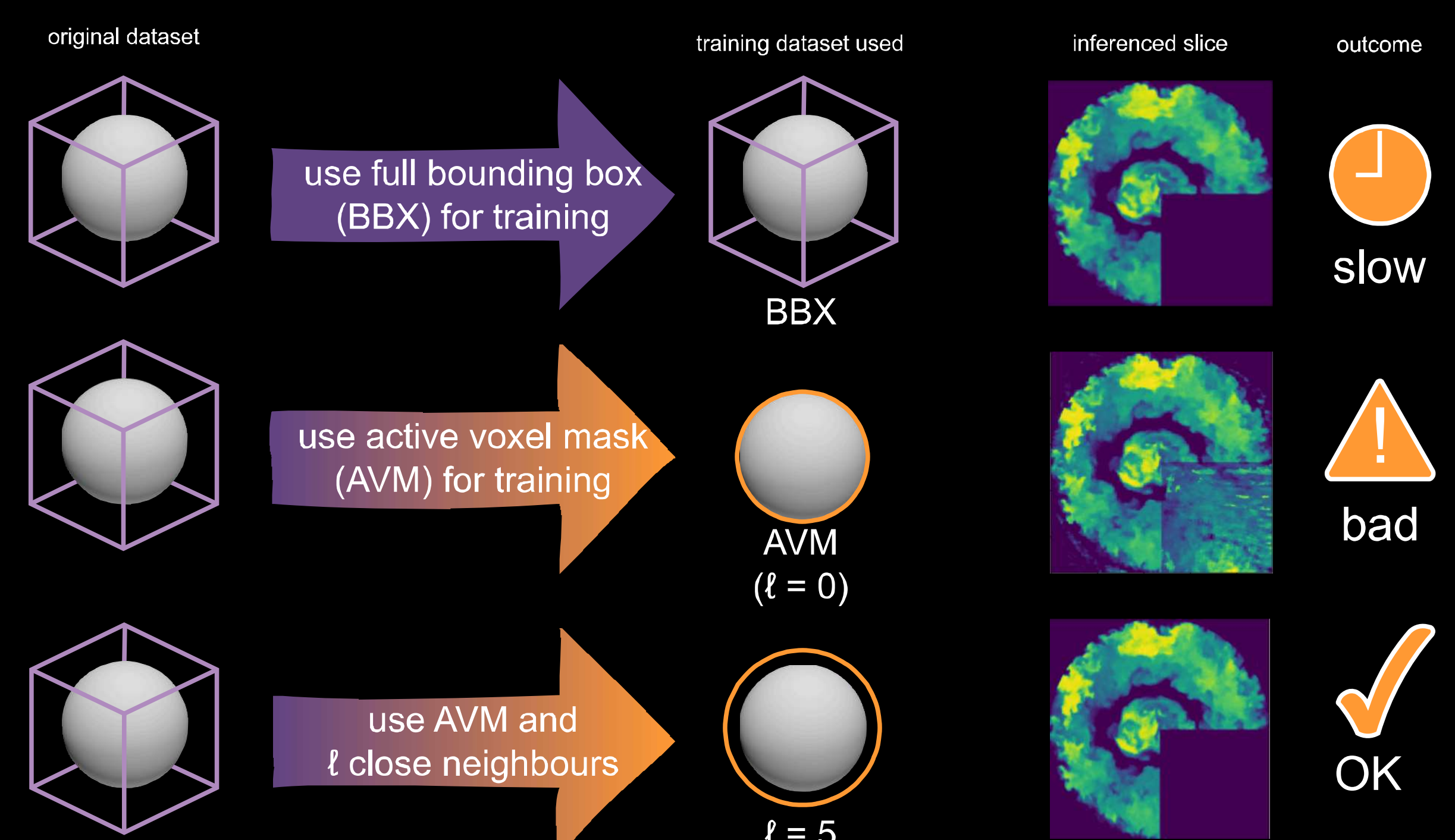
Implications & takeaways

Structure free INRs offer a simple and scalable way to compress volumetric data. Fourier encoding enables high-fidelity reconstructions with compact networks. Selective voxel sampling reduces compute without sacrificing quality, making INRs faster and more practical for real world visualization workloads.

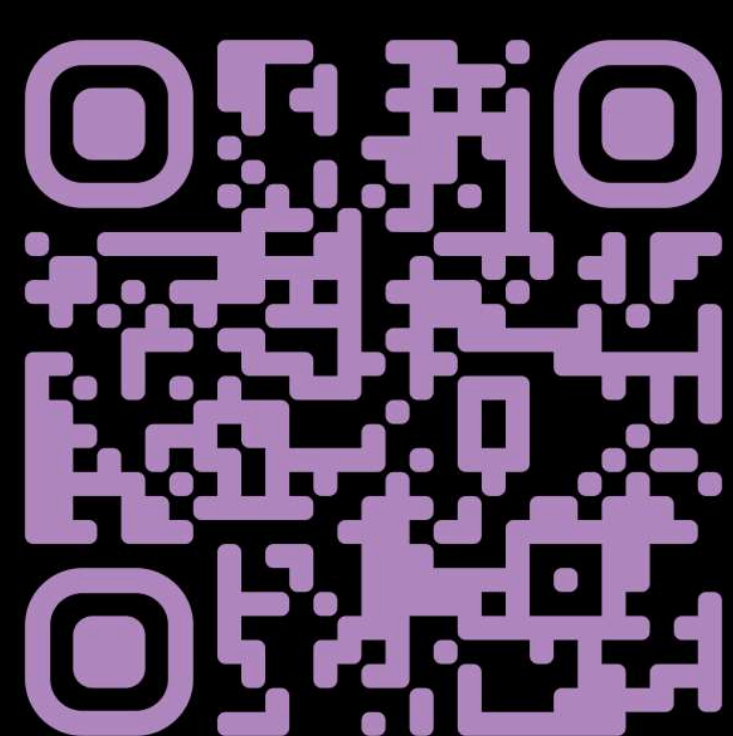
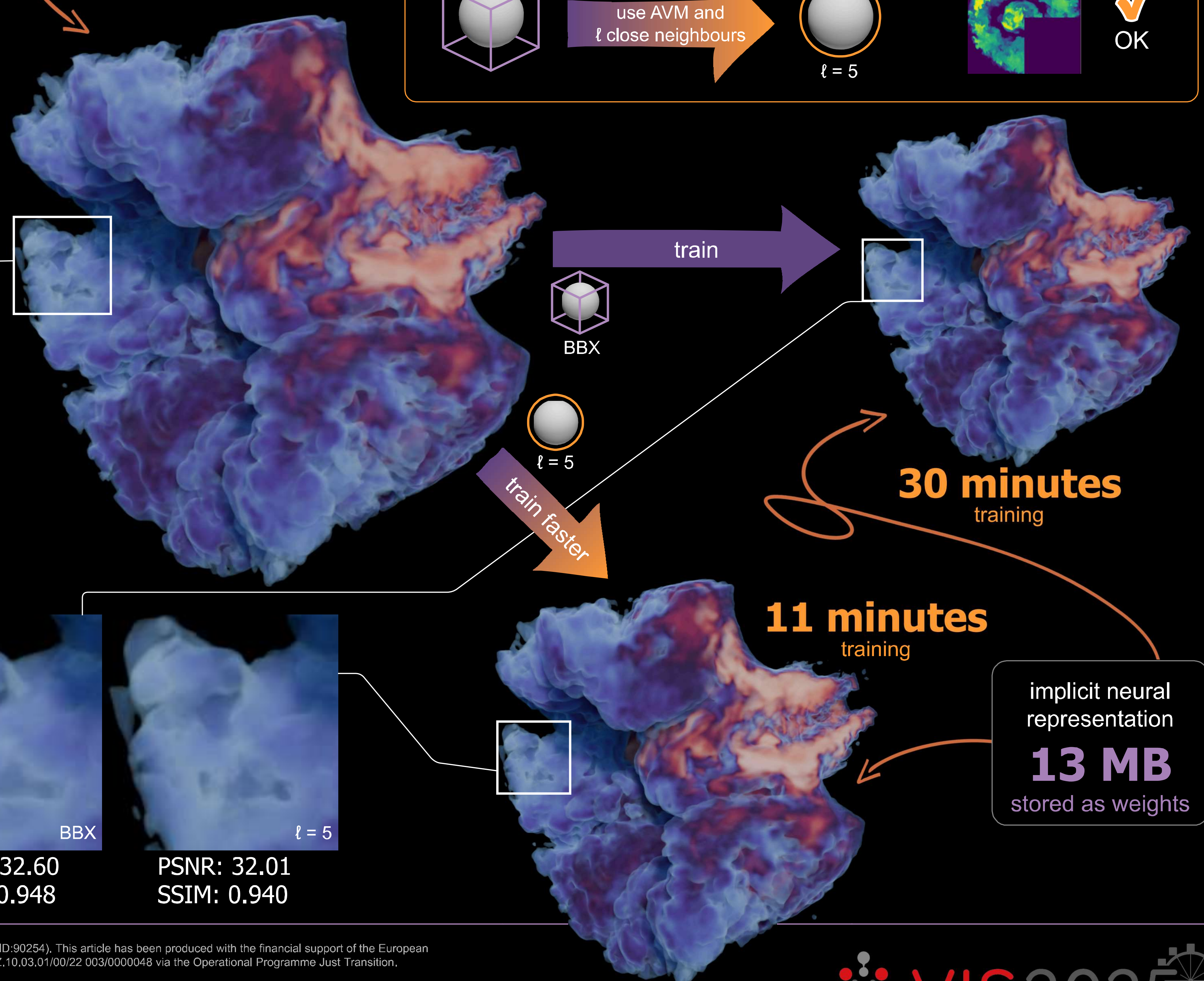
Fourier features to capture details



Dilation size (ℓ)



original data
125 MB
stored as values



Preprint at
ArXiv.org



Reference



PSNR: 32.60
SSIM: 0.948



PSNR: 32.01
SSIM: 0.940